

When are Imprecise Bandits Computationally Tractable?

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1. Setting

An instance of the imprecise bandits problem is a tuple $\mathcal{M} = (X, D, H, r)$, where X is a set of arms, D is a set of possible outcomes, H is a set of mappings of type $X \rightarrow \square D$ where $\square D$ denotes the set of credal sets over D , and $r : X \times D \rightarrow \mathbb{R}$ is a reward function.

The imprecise bandits game is played between a learner and an adversary for T rounds (both adversary and learner know T). At the beginning of the game, the adversary picks some $h^* \in H$. The learner knows X, D, H, r , and T , but does not have knowledge of h^* . Then, at each round t :

1. Learner picks an arm $x_t \in X$.
2. Adversary picks a distribution $P_t \in h^*(x_t)$.
3. An outcome $y_t \sim P_t$ is drawn and shown to the learner as feedback.
4. Learner gets reward $r(x_t, y_t)$.

Given a hypothesis $h \in H$, we define the value functions:

$$v_h(x) := \min_{P \in h(x)} \mathbb{E}_{y \sim P}[r(x, y)], \quad V_h := \max_{x \in X} v_h(x).$$

We use the shorthand $v^*(x)$ and V^* to denote the value functions corresponding to the true hypothesis h^* .

The regret over horizon T is

$$R_T := TV^* - \mathbb{E} \left[\sum_{t=1}^T r(x_t, y_t) \right].$$

Here the expectation in the second term is taken wrt the randomness in the learner's and the adversary's strategies as well as the randomness arising from sampling $y_t \sim P_t$.

We are interested in learner strategies that can achieve a sublinear (in T) regret for all true hypotheses $h^* \in H$. We say that an imprecise bandits instance $\mathcal{M} = (X, D, H, r)$ has a *statistically efficient* learner if such a learner strategy exists.

Computational tractability. In this paper, we are interested in studying the computational complexity of the learner. A *learner policy* π is a randomized algorithm that takes as input a history $h_t = ((x_i, y_i))_{i=1}^t$ and the description of the problem instance $\mathcal{M}$ and returns a next arm x_{t+1} . The computational complexity of the problem depends crucially on the exact nature of access the policy is given to $\mathcal{M}$. For example, on one extreme, one can consider a non-uniform access such that a policy $\pi^{\mathcal{M}}$ is merely indexed by the instance $\mathcal{M}$ as opposed to $\mathcal{M}$ being an input to a computational procedure. However, this makes several instances rather trivial. In this paper we mostly consider specific encodings of $\mathcal{M}$. We also occasionally assume π to have oracle access to $\mathcal{M}$ with certain specific oracles. We will make these assumptions clear in the respective sections.

We will say that a learner is *computationally efficient* if it runs in time polynomial in the size of its input and achieves a regret $R_T \leq \text{poly}(|\mathcal{M}|)T^{1-\alpha}$ for some $\alpha > 0$.

1.1. Linear Imprecise Bandits

For most of this work, we study the following linear specialization of the instance $\mathcal{M} = (X, D, H, r)$ defined above. Let $X \subseteq \mathbb{R}^{d_X}$ be the arm set, and let $D \subseteq \mathbb{R}^{d_D}$ be the outcome set. We assume that X and D are Euclidean balls of radii R_X and R_D (not necessarily centred at the origin). The reward is affine in the arm and outcome:

$$r(x, y) := a^T x + b^T y + c.$$

Let the hypothesis class H be parametrized by $Z \subseteq \mathbb{R}^{d_Z}$. For every $z \in Z$, we define the hypothesis h_z by first defining the possible expected values of the distributions that h_z is allowed to choose. In particular, let

$$K_z(x) := \{y \in D : C_z y + B_z x + d_z = 0\}. \quad (1)$$

Here B_z , C_z , and d_z are z -dependent matrices and vectors of appropriate dimensions. Now, hypothesis $h_z : X \rightarrow \square D$ is given by

$$h_z(x) := \{P \in \Delta D : \mathbb{E}_{y \sim P}[y] \in K_z(x)\}.$$

Let $z^* \in Z$ represent the true hypothesis, so that $h^* = h_{z^*}$. The true feasible mean set is $K^*(x) := K_{z^*}(x)$.

We also assume a uniform non-tangency condition: there is a constant $S \in (0, 1]$ such that, for every $x \in X$ and every $p \in \mathbb{R}^{d_D}$ satisfying $C_{z^*} p + B_{z^*} x + d_{z^*} = 0$ with $p \notin D$,

$$\text{dist}(p, D) \geq S \text{dist}(p, K^*(x)).$$

For a Euclidean ball, this says that the affine solution spaces defined by the true constraints remain uniformly bounded away from tangency to D . This is the transversality condition used in [1].

Because r is affine in its outcome argument, the value functions defined above satisfy

$$v^*(x) = \min_{P \in h^*(x)} \mathbb{E}_{y \sim P}[r(x, y)] = \min_{y \in K^*(x)} r(x, y), \quad V^* := \max_{x \in X} v^*(x).$$

The regret over horizon T is

$$R_T := TV^* - \mathbb{E} \left[\sum_{t=1}^T r(x_t, y_t) \right].$$

We measure computation in the arithmetic and weak-optimization model used below, in which the coordinates of the observed outcomes are real inputs. The model-oblivious policy constructed below is **polynomial-time** if one fixed polynomial in L_{pub} and T bounds its total running time through round T , including the cost and requested accuracy of its weak-optimization calls. Since the horizon is encoded as 1^T , this is polynomial in the length of the reduced input and observed transcript. The finite-precision implementation is given in the computational tractability subsection.

Problem 1. Find a polynomial-time policy π , a fixed constant $\varepsilon > 0$, and a fixed polynomial P such that, for every linearly presented instance $\mathcal{M}$ whose reduced part has a rational description $\mathcal{P}$, every horizon $T \geq 1$, every valid realization (z^*, S) inducing h^* , and every adaptive nature policy ν compatible with h^* ,

$$R_T \leq P(d_X, d_D, L_{\text{pub}}, R_X, R_D, \|b\|_2, C_r, S^{-1}) T^{1-\varepsilon}.$$

The expectation in R_T is over the private randomization of π and the outcomes sampled by ν .

Theorem 1 (efficient $T^{\frac{2}{3}}$ learning). There is a horizon-aware policy for Problem 1 whose arithmetic and weak-optimization running time is polynomial in T and the reduced-description length and which, for every true hypothesis satisfying the uniform non-tangency condition and every compatible adaptive nature policy, satisfies

$$R_T \leq \tilde{O}\left(P(d_X, d_D, R_X, R_D, \|b\|_2, C_r, S^{-1}) T^{\frac{2}{3}}\right),$$

where P is a fixed polynomial. Apart from the uniform non-tangency condition on the true feasible sets, no geometric property of H or its parameter set Z is used.

Since every fixed polylogarithmic factor is $O(T^{\frac{1}{12}})$, this theorem solves Problem 1 with, for example, $\varepsilon = \frac{1}{4}$.

2. Warmup: Noiseless Setting

In this section, playing an arm x reveals an exact feasible response $y \in K^*(x)$; there is no sampling noise. However, there still is the Knightian uncertainty corresponding to the adversary picking an arbitrary outcome inside $K^*(x)$.

Let $N := \ker C_{z^*}$.

Lemma 1. There exists an affine map $f_z : X \rightarrow \mathbb{R}^{d_D}$ and a linear subspace N_z independent of x such that

$$f_z(x) + N_z = \{y \in \mathbb{R}^{d_D} : C_z y + B_z x + d_z = 0\}, \quad x \in X.$$

Consequently, there exists an f and N such that

$$K^*(x) = (f(x) + N) \cap D, \quad x \in X.$$

Proof. Fix x and let $y_1, y_2 \in K_z(x)$. Then $C_z(y_1 - y_2) = 0$, which means $y_1 - y_2$ must lie in a subspace determined by z . We can pick some $y \in K_z(x)$ arbitrarily and set $f_z(x) = y$. This is clearly an affine mapping. ■

Note that as the proof demonstrates, f_z need not be unique. Indeed, for each z there can be multiple f_z 's that satisfy the requirement of the lemma. However, each z determines a unique N_z .

Lemma 2. Suppose that, for every arm $x \in X$, we have identified a nonempty set $K'(x) \subseteq K^*(x)$, and suppose that whenever the learner plays x , nature chooses an outcome in $K'(x)$. Define

$$v'(x) := \min_{y \in K'(x)} r(x, y),$$

and choose

$$x' \in \operatorname{argmax}_{x \in X} v'(x).$$

Then playing x' on every round incurs nonpositive regret with respect to the original robust benchmark, i.e. $R_T \leq 0$.

Lemma 3. There exist $d_X + 1$ arms $x^{(0)}, \dots, x^{(d_X)} \in X$ such that, from any possible sequence of responses $y^{(i)} \in K^*(x^{(i)})$, we can compute an affine map $\hat{f}$ such that

$$K^*(x) = (\hat{f}(x) + N) \cap D, \quad x \in X.$$

Proof. Let f be any affine map supplied by Lemma 1. Let $x^{(0)}, \dots, x^{(d_X)}$ be any affine basis of X . Play them in sequence and let $y^{(0)}, \dots, y^{(d_X)}$ be the corresponding outcomes chosen by the adversary. Let $\hat{f}$ be the unique affine interpolator for which $\hat{f}(x^{(i)}) = y^{(i)}$ for all i . Since $y^{(i)} \in K^*(x^{(i)})$, we have that $\hat{f}(x^{(i)}) + N = f(x^{(i)}) + N$ for all i . Since the $x^{(i)}$ form an affine basis, $\hat{f}(x) + N = f(x) + N$ for all $x \in X$. ■

Algorithm 1: Noiseless imprecise bandit.

1. Play the $d_X + 1$ arms from Lemma 3 and use their responses to construct $\hat{f}$.
2. Initialize $N_{d_X+2} := \{0\}$.
3. For each round $t = d_X + 2, \dots, T$:
 - For every $x \in X$, define the provisional feasible set

$$K'_t(x) := (\hat{f}(x) + N_t) \cap D.$$
 - If $K'_t(x) = \emptyset$ for some $x \in X$, choose any such arm as x_t . Otherwise, choose

$$x_t \in \operatorname{argmax}_{x \in X} \min_{y \in K'_t(x)} r(x, y).$$
 - Play x_t and observe y_t .
 - If $y_t - \hat{f}(x_t) \notin N_t$, set

$$N_{t+1} := \operatorname{span}(N_t \cup \{y_t - \hat{f}(x_t)\}).$$

Otherwise, set $N_{t+1} := N_t$.

Every N_t maintained by Algorithm 1 is a subspace of N . Whenever it is updated, its dimension increases by one. Moreover, if a provisional feasible set is empty, playing an arm with an empty provisional set necessarily triggers such an update. There can therefore be at most $\dim(N) \leq d_D$ update rounds.

On every remaining round, all the provisional feasible sets are nonempty and the observed response belongs to $K'_t(x_t)$. Hence Lemma 2 shows that the reward on that round is at least the original robust benchmark. Charging at most a constant regret to each of the $d_X + 1$ initial rounds and to each update round gives $R_T \leq O(d_X + d_D)$.

2.1. Computational Tractability

All the steps in Algorithm 1 can be carried out in polynomial time. First, affine interpolation can be done via Gaussian elimination.

The subspace N_t can be stored through an orthonormal basis. Given the new residual

$$h_t := y_t - \hat{f}(x_t),$$

project h_t onto $N_t^\perp$. If the projection is zero, the span does not change. Otherwise, normalize the projection and append it to the stored basis. This is an incremental Gram-Schmidt and takes polynomial time.

It remains to compute the arm in the planning step.

Lemma 4. *Fix a round t . Given rational descriptions of $X, D, \hat{f}$, and N_t , one can compute, in polynomially many arithmetic operations, an arm $x_{\text{emp}} \in X$ such that, if $K'_t(x) = \emptyset$ for some $x \in X$, then $K'_t(x_{\text{emp}}) = \emptyset$.*

Proof. Clearly, for a given x , the set $K'_t(x)$ is empty if and only if the distance between c_D and $\hat{f}(x) + N_t$ is bigger than R_D . Thus to decide if there exists an $x \in X$ for which $K'_t(x)$ is empty, we need to find the arm that maximizes the distance between c_D and $\hat{f}(x) + N_t$. To see why this can be done in polynomial time, let $q_t(x)$ be the point in $\hat{f}(x) + N_t$ that is closest to c_D . Note that $q_t(x)$ can be written as an affine function.

$$q_t(x) := c_D + (I - P_t)(\hat{f}(x) - c_D).$$

Here P_t is the projection on N_t .

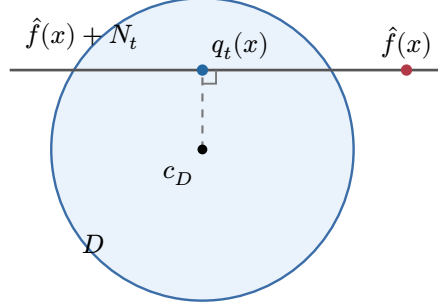


Figure 1: Two-dimensional outcome geometry when N_t is one-dimensional. The dashed segment is orthogonal to $\hat{f}(x) + N_t$, so its endpoint is $q_t(x)$, the point on that line closest to c_D .

Now we need to compute

$$x_{\text{emp}} \in \operatorname{argmax}_{x \in X} \|q_t(x) - c_D\|_2^2.$$

This is a quadratic optimization problem over the Euclidean ball X , which is polynomial-time solvable using the result in [2]. ■

Lemma 5. *Fix a round t and suppose that $K'_t(x)$ is nonempty for every $x \in X$. Let the rational problem data have total bit length L . For every $\varepsilon \in (0, 1)$, one can compute an arm $x_\varepsilon \in X$ in time $\text{poly}(L, \log(\frac{1}{\varepsilon}))$ such that*

$$\min_{y \in K'_t(x_\varepsilon)} r(x_\varepsilon, y) \geq \max_{x \in X} \min_{y \in K'_t(x)} r(x, y) - \varepsilon.$$

Proof. Compute the orthogonal projector P_t onto N_t as in the proof of Lemma 4, and define

$$q_t(x) := c_D + (I - P_t)(\hat{f}(x) - c_D).$$

Since every provisional set is nonempty, Pythagoras' theorem gives

$$K'_t(x) = q_t(x) + \left\{ u \in N_t : \|u\|_2 \leq \sqrt{R_D^2 - \|q_t(x) - c_D\|_2^2} \right\}.$$

Put $\beta_t := \|P_t b\|_2$. For a fixed arm x , minimizing the linear reward over the displayed ball moves in direction $-P_t b$, and hence

$$v'_t(x) := \min_{y \in K'_t(x)} r(x, y) = a^T x + b^T q_t(x) + c - \beta_t \sqrt{R_D^2 - \|q_t(x) - c_D\|_2^2}.$$

Introduce a scalar $s \geq 0$. Maximizing v'_t is equivalent to the QCQP

$$\max_{x, s} a^T x + b^T q_t(x) + c - \beta_t s$$

subject to

$$\|x - c_X\|_2^2 \leq R_X^2, \quad s^2 + \|q_t(x) - c_D\|_2^2 = R_D^2, \quad s \geq 0,$$

where c_X is the centre of X . Represent the equality by two quadratic inequalities. If $R_X, R_D > 0$, add the redundant constraint

$$\frac{\|x - c_X\|_2^2}{R_X^2} + \frac{s^2}{R_D^2} \leq 2.$$

This is an ellipsoid in the joint variable (x, s) . Thus the QCQP has a fixed number of quadratic constraints, one of which is ellipsoidal. Bienstock's theorem [3] returns, in time polynomial in L and $\log(\frac{1}{\delta})$, a solution (x^*, s^*) such that for each constraint $g_i(x, s) \leq 0$, we have that $g_i(x^*, s^*) \leq \delta$, and the value of the objective function $\Phi_t(x, s) := a^T x + b^T q_t(x) + c - \beta_t s$ at (x^*, s^*) is δ -close to the optimal, i.e., for every feasible (x, s) , $\Phi_t(x^*, s^*) \geq \Phi_t(x, s) - \delta$.

The output x^* of Bienstock's algorithm may not be a feasible arm. But to obtain a feasible arm, we can simply project x^* onto X to obtain x_ε . It is easy to see that this does not change the value of the objective function too much. Indeed, set $s_\varepsilon := \sqrt{R_D^2 - \|q_t(x_\varepsilon) - c_D\|_2^2}$. The projection moves x^* by $O(\sqrt{\delta})$, and thus moves s^* by at most $O(\delta^{\frac{1}{4}})$. Consequently the linear objective changes by at most $C\delta^{\frac{1}{4}}$, where $C \geq 1$ is a constant. Choose $\delta \leq (\frac{\varepsilon}{2C})^4$. The resulting arm is feasible and its robust value is within ε of the optimum. Since $\log(\frac{1}{\delta}) = \text{poly}(L) + O(\log(\frac{1}{\varepsilon}))$, the running time is polynomial. ■

Set $\varepsilon = T^{-2}$ in Lemma 5. On each round, the computed arm then loses at most T^{-2} relative to the exact planning solution, so approximate planning contributes at most T^{-1} to total regret.

3. Noisy Setting

In this section, constants hidden by O and $\tilde{O}$ may depend on the fixed problem parameters. In addition, $\tilde{O}$ suppresses logarithmic factors in T and $\frac{1}{\delta}$.

In the noisy setting, estimating the true N is tricky. To reduce the level of noise, we can work with average outcomes over blocks of length n instead of each individual outcome. Suppose m_i is the conditional mean corresponding to the outcome y_i . Then over a block of size n , we can bound $\|\frac{1}{n} \sum y_i - \frac{1}{n} \sum m_i\| \leq O(n^{-\frac{1}{2}})$ with high probability. Due to the convexity of D , $\frac{1}{n} \sum m_i$ also lies in D , and therefore, the "residuals" calculated using $\bar{y} = \frac{1}{n} \sum y_i$ should be close to the true residuals. However, it is still risky to compute the subspace containing this residual and committing to it for the future steps. A small error in the direction of the calculated subspace can lead to large errors further out. Thus to get better control over the estimation errors, we approximate the subspace N with an ellipsoid. We start by computing an approximate $\hat{f}$ using an affine basis, and assume that N is a unit ball of radius λ where $\lambda \leq O(n^{-\frac{1}{2}})$. Then if a residual arrives that lies outside the current ellipsoid, we expand the ellipsoid in a specific way along the direction of the residual.

Lemma 6. Fix $\delta \in (0, 1)$ and $1 \leq n \leq T$. There exist $d_X + 1$ affinely independent anchor arms such that, after playing each arm for n rounds and interpolating their empirical average outcomes, the resulting affine map $\hat{f}$ satisfies the following with probability at least $1 - \delta$. There is an affine map f and a linear subspace N such that $K^*(x) = (f(x) + N) \cap D$ and

$$\sup_{x \in X} \|\hat{f}(x) - f(x)\|_2 \leq \tilde{O}(n^{-\frac{1}{2}}).$$

Proof. Let $x^{(0)}, \dots, x^{(d_X)}$ be any affine basis of X . Play each arm for n rounds. Let $\bar{y}^{(i)}$ be the average observed outcome for arm i , and let $\bar{m}^{(i)}$ be the corresponding average conditional mean. Since $K^*(x^{(i)})$ is convex, $\bar{m}^{(i)} \in K^*(x^{(i)})$. Let $\hat{f}$ and f be the unique affine interpolators satisfying $\hat{f}(x^{(i)}) = \bar{y}^{(i)}$ and $f(x^{(i)}) = \bar{m}^{(i)}$ for every i . By Lemma 3, $K^*(x) = (f(x) + N) \cap D$ for every $x \in X$.

Within each block, $y_t - m_t$ is a martingale-difference sequence. Since D is bounded, the Azuma-Hoeffding inequality for bounded martingale differences (see [4], [5]), together with a union bound over the finitely many anchors and outcome coordinates, implies that, with probability at least $1 - \delta$,

$$\max_{0 \leq i \leq d_X} \|\bar{y}^{(i)} - \bar{m}^{(i)}\|_2 \leq \tilde{O}(n^{-\frac{1}{2}}).$$

Now consider an arbitrary $x \in X$, and let $\alpha_0, \dots, \alpha_{d_X}$ be its affine coordinates with respect to the chosen basis. Since X is compact, there is a constant L , independent of x , such that

$$\sum_{i=0}^{d_X} |\alpha_i| \leq L.$$

Since $\hat{f}$ and f are affine and agree with the corresponding values at the anchors,

$$\|\hat{f}(x) - f(x)\|_2 \leq \sum_{i=0}^{d_X} |\alpha_i| \|\bar{y}^{(i)} - \bar{m}^{(i)}\|_2 \leq L \tilde{O}(n^{-\frac{1}{2}}).$$

Taking the supremum over $x \in X$ proves the result. ■

In the noisy setting, taking the span of the observed residuals is unstable: even a small amount of noise can introduce a spurious direction. We therefore replace the subspace N_t by a learned residual ellipsoid.

Fix $\delta \in (0, 1)$ and $1 \leq n \leq T$, and let $\hat{f}$ and f be as in Lemma 6. One can choose $\lambda = \tilde{O}(n^{-\frac{1}{2}})$ with $\lambda \geq n^{-\frac{1}{2}}$ so that, conditional on the anchor conclusion, the conclusions of the following two lemmas hold simultaneously throughout the horizon with probability at least $1 - \delta$.

Algorithm 2: Noisy imprecise bandit (n, λ) .

Parameters: block length $n \in \mathbb{N}$ and ridge parameter $\lambda > 0$.

1. Choose the $d_X + 1$ anchor arms from Lemma 6. Play each anchor for n consecutive rounds, compute its empirical average $\bar{y}^{(i)}$, and construct the affine interpolant $\hat{f}$. If the horizon ends during this phase, stop.
2. Initialize $M := 0$ and let $\hat{G}_0$ be the matrix with no columns.
3. At the beginning of each subsequent block, define

$$V_M := \lambda^2 I_{d_D} + \hat{G}_M \hat{G}_M^T \in \mathbb{R}^{d_D \times d_D}, \quad \mathcal{E}_M := \{u \in \mathbb{R}^{d_D} : u^T V_M^{-1} u \leq 1\}.$$

For every arm $x \in X$, define

$$\hat{K}_M(x) := (\hat{f}(x) + \mathcal{E}_M) \cap D.$$

4. If $\hat{K}_M(x) = \emptyset$ for some $x \in X$, choose any such arm. Otherwise, choose

$$x \in \operatorname{argmax}_{x' \in X} \min_{y \in \hat{K}_M(x')} r(x', y).$$
5. Play the chosen arm for n rounds. If fewer than n rounds remain, play until the horizon and stop without updating. Otherwise, let $\bar{y}$ be the block-average outcome and set

$$\hat{g} := \bar{y} - \hat{f}(x).$$

6. If $\hat{g} \notin \mathcal{E}_M$, declare the block informative, set $\hat{G}_{M+1} := [\hat{G}_M, \hat{g}]$, and then increase M by one. Otherwise leave $\hat{G}_M$ and M unchanged. Return to Step 3 and continue until time T .
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Lemma 7. *At most $O(\log T)$ blocks of Algorithm 2 are informative.*

Proof. Enumerate the stored residuals as $\hat{g}_1, \dots, \hat{g}_M$. After j of them have been stored,

$$V_j := \lambda^2 I_{d_D} + \sum_{i=1}^j \hat{g}_i \hat{g}_i^T.$$

Every block average lies in D . The anchor averages also lie in D , and affine interpolation from the fixed anchors has uniformly bounded coefficients on the compact set X . Hence $\|\hat{g}_j\|_2 = O(1)$ for every block.

When $\hat{g}_j$ is stored, the block is informative, so

$$\hat{g}_j^T V_{j-1}^{-1} \hat{g}_j > 1.$$

The matrix determinant lemma [6] therefore gives

$$\det(V_j) = \det(V_{j-1}) (1 + \hat{g}_j^T V_{j-1}^{-1} \hat{g}_j) > 2 \det(V_{j-1}).$$

Since $V_0 = \lambda^2 I_{d_D}$, iteration yields

$$\det(V_M) > 2^M \lambda^{2d_D}.$$

On the other hand, $M \leq \frac{T}{n}$, and therefore

$$\operatorname{tr}(V_M) = d_D \lambda^2 + \sum_{j=1}^M \|\hat{g}_j\|_2^2 \leq d_D \lambda^2 + O\left(\frac{T}{n}\right).$$

Applying the arithmetic–geometric mean inequality to the eigenvalues of V_M gives

$$\det(V_M) \leq \left(\lambda^2 + O\left(\frac{T}{n}\right) \right)^{d_D}.$$

Combining the two determinant bounds,

$$M \leq d_D \log_2 \left(1 + O\left(\frac{T}{n\lambda^2}\right) \right) = O(\log T),$$

where the last equality uses $\lambda \geq n^{-\frac{1}{2}}$. ■

Lemma 8. *Let $\hat{f}$ be the empirical affine interpolant constructed during the anchor phase of Algorithm 2, and let f be the affine interpolant of the corresponding average conditional means, as in Lemma 6. For every current ellipsoid $\mathcal{E}_M$, every $x \in X$, and every $u \in \mathcal{E}_M$,*

$$\text{dist}(\hat{f}(x) + u, f(x) + N) \leq \tilde{O}(n^{-\frac{1}{2}}).$$

Proof. We prove by induction on M .

We first show that this bound holds for the initial matrix $V_0 = \lambda^2 I_{d_D}$, and then show that it remains true after every update from V_M to V_{M+1} .

For the initial matrix $V_0 = \lambda^2 I_{d_D}$, the ellipsoid $\mathcal{E}_0$ is the Euclidean ball of radius λ . By Lemma 6, there exists some $\beta_n \leq \tilde{O}(n^{-\frac{1}{2}})$ such that $\|\hat{f}(x) - f(x)\|_2 \leq \beta_n$ for every $x \in X$. Therefore, by the triangle inequality, for every $x \in X$ and $u \in \mathcal{E}_0$,

$$\text{dist}(\hat{f}(x) + u, f(x) + N) \leq \|\hat{f}(x) - f(x)\|_2 + \|u\|_2 \leq \beta_n + \lambda \leq \tilde{O}(n^{-\frac{1}{2}}).$$

Thus the claimed bound holds when $M = 0$.

Geometrically, $\hat{f}(x) + \mathcal{E}_0$ is a ball of radius λ centred at $\hat{f}(x)$, while $f(x) + N$ is the affine space that we want this ball to remain close to. Each informative update adds a new generating direction $\hat{g}$ to the ellipsoid. The component of $\hat{g}$ lying in N only moves the ellipsoid parallel to $f(x) + N$ and therefore does not increase the distance from that affine space. Only the component orthogonal to N matters. Thus, if $\hat{g}$ is within η_n of N , one update can increase the distance by at most η_n . An uninformative block does not change the ellipsoid at all.

To make this precise, consider a complete block $\mathcal{B}$ at arm x and let

$$\bar{m} := \frac{1}{n} \sum_{t \in \mathcal{B}} m_t.$$

Since $\bar{m} \in K^*(x)$, the vector $g := \bar{m} - f(x)$ belongs to N . The empirical residual is $\hat{g} := \bar{y} - \hat{f}(x)$. Conditionally on the history at the beginning of any adaptively selected block, the vectors $y_t - m_t$ remain martingale differences. The same concentration argument as in Lemma 6, followed by a union bound over the at most $\frac{T}{n}$ complete blocks, therefore gives, simultaneously for every block,

$$\|\bar{y} - \bar{m}\|_2 \leq \tilde{O}(n^{-\frac{1}{2}}).$$

Combining this with the uniform anchor-interpolation bound gives

$$\|\hat{g} - g\|_2 \leq \tilde{O}(n^{-\frac{1}{2}}).$$

We may therefore choose $\eta_n = \tilde{O}(n^{-\frac{1}{2}})$ so that every stored residual satisfies $\text{dist}(\hat{g}, N) \leq \eta_n$. We prove the following more explicit form of the induction claim:

$$\text{dist}(\hat{f}(x) + u, f(x) + N) \leq \beta_n + \lambda + M\eta_n, \quad x \in X, u \in \mathcal{E}_M.$$

The base case above establishes this claim when $M = 0$.

Now consider an update from V_M to V_{M+1} . Suppose that the bound holds for M and that the informative block producing the update has residual $\hat{g}$. Let $Q_M := [\hat{G}_M, \lambda I_{d_D}]$. Since $Q_M Q_M^T = V_M$ and Q_M has full row rank,

$$\mathcal{E}_M = \{Q_M \theta : \|\theta\|_2 \leq 1\}.$$

After storing $\hat{g}$, every $u \in \mathcal{E}_{M+1}$ can therefore be written as

$$u = Q_M \theta + a \hat{g}, \quad \|\theta\|_2^2 + a^2 \leq 1.$$

Set $u_0 := Q_M \theta$. Then $u_0 \in \mathcal{E}_M$, and since N is a linear subspace, for every $x \in X$,

$$\text{dist}(\hat{f}(x) + u, f(x) + N) \leq \text{dist}(\hat{f}(x) + u_0, f(x) + N) + |a| \text{dist}(\hat{g}, N) \leq \beta_n + \lambda + (M+1)\eta_n.$$

This proves the induction claim. By Lemma 7, $M = O(\log T)$, and therefore

$$\beta_n + \lambda + M\eta_n = \tilde{O}(n^{-\frac{1}{2}}).$$

■

Lemma 9. *Under the uniform non-tangency condition, for every $x \in X$ and every $y \in \mathbb{R}^{d_D}$,*

$$\text{dist}(y, K^*(x)) \leq (1 + S^{-1}) \text{dist}(y, f(x) + N) + S^{-1} \text{dist}(y, D).$$

In particular, if $y \in D$, then

$$\text{dist}(y, K^*(x)) \leq (1 + S^{-1}) \text{dist}(y, f(x) + N).$$

Proof. Let p be the Euclidean projection of y onto $f(x) + N$. If $p \in D$, then $p \in K^*(x)$ and the result is immediate. Otherwise, the uniform non-tangency condition and the triangle inequality give

$$\text{dist}(p, K^*(x)) \leq S^{-1} \text{dist}(p, D) \leq S^{-1}(\|y - p\|_2 + \text{dist}(y, D)).$$

Since $\|y - p\|_2 = \text{dist}(y, f(x) + N)$, one more application of the triangle inequality proves the first claim. The second follows by setting $\text{dist}(y, D) = 0$. ■

Lemma 10. *Suppose that the conclusions of Lemma 6 and Lemma 8 hold. Let $\mathcal{B}$ be a complete n -round block that is declared uninformative, and let x be the arm played in that block. Then its realized average regret satisfies*

$$\frac{1}{n} \sum_{t \in \mathcal{B}} (V^* - r(x, y_t)) \leq \tilde{O}(n^{-\frac{1}{2}}).$$

Proof. Let M be the state at the start of the block and let

$$\bar{y} := \frac{1}{n} \sum_{t \in \mathcal{B}} y_t.$$

Since the block is uninformative, $\hat{g} := \bar{y} - \hat{f}(x)$ belongs to $\mathcal{E}_M$.

Since D is convex, $\bar{y} \in D$. Therefore,

$$\bar{y} \in (\hat{f}(x) + \mathcal{E}_M) \cap D = \hat{K}_M(x).$$

In particular, this block could not have been selected by the empty-set branch of Step 4, so its arm was selected by the planning rule. Define

$$\hat{v}_M(x') := \min_{y \in \hat{K}_M(x')} r(x', y).$$

For every $x' \in X$ and $y \in \hat{K}_M(x')$, Lemma 8 gives

$$\text{dist}(y, f(x') + N) \leq \tilde{O}(n^{-\frac{1}{2}}).$$

Since $y \in D$, applying Lemma 9 gives

$$\text{dist}(y, K^*(x')) \leq \tilde{O}(n^{-\frac{1}{2}}).$$

Since the reward is Lipschitz in its outcome argument,

$$\hat{v}_M(x') \geq v^*(x') - \tilde{O}(n^{-\frac{1}{2}}).$$

Let x^* maximize v^* . The planning rule and the fact that $\bar{y} \in \hat{K}_M(x)$ now give

$$r(x, \bar{y}) \geq \hat{v}_M(x) \geq \hat{v}_M(x^*) \geq V^* - \tilde{O}(n^{-\frac{1}{2}}).$$

Finally, the reward is affine in the outcome and the arm is fixed throughout the block, so

$$r(x, \bar{y}) = \frac{1}{n} \sum_{t \in \mathcal{B}} r(x, y_t).$$

This proves the average-regret bound. ■

Taking $\delta = (T + 1)^{-2}$, $n = \lceil T^{\frac{2}{3}} \rceil$, and λ chosen as above now gives the claimed regret rate. The anchor phase, the $O(\log T)$ informative blocks from Lemma 7, and the final partial block contribute $\tilde{O}(n)$ regret. By Lemma 10, all remaining rounds contribute $\tilde{O}(Tn^{-\frac{1}{2}})$. Hence, when the anchor conclusion and the joint ridge conclusion both hold,

$$R_T \leq \tilde{O}(n + Tn^{-\frac{1}{2}}) = \tilde{O}(T^{\frac{2}{3}}).$$

These conclusions fail with probability at most $2(T + 1)^{-2}$, and $R_T \leq C_r T$ always, so taking expectations only adds $O(\frac{C_r}{T})$.

3.1. Computational Tractability

The statistical argument above presents Algorithm 2 using exact emptiness tests and exact maximization over the single provisional set $\hat{K}_M(x)$. We now explain how to implement these operations with weak

finite-precision optimization. The additional sets introduced in this section provide numerical slack only; they do not change the statistical idea.

The matrix updates, leverage calculations, and affine interpolation are standard polynomial-time linear algebra. Moreover, $V_M - \lambda^2 I_{d_D}$ is positive semidefinite, while $\lambda \geq T^{-\frac{1}{2}}$ and the stored residuals have bounded norm. Hence all inversions can be carried out to the required inverse-polynomial precision in polynomial time. For this implementation, one may choose the centre c_X of X and the arms $c_X + R_X e_i$, $i = 1, \dots, d_X$, as the affine basis. Their interpolation coefficients and the resulting affine map are computed by Gaussian elimination and matrix multiplication, with polynomially controlled bit complexity. This conditioning detail is irrelevant to the dependence on n and was therefore omitted from the statistical argument.

3.1.1. Empty-Fiber Certification

Fix $\rho := \lambda$, let c_D be the centre of D , and define

$$D_1 := \{y : \|y - c_D\|_2 \leq R_D + \rho\}, \quad D_2 := \{y : \|y - c_D\|_2 \leq R_D + 2\rho\},$$

$$\hat{K}_M^{\text{in}}(x) := (\hat{f}(x) + 2\mathcal{E}_M) \cap D_1, \quad \hat{K}_M^{\text{out}}(x) := (\hat{f}(x) + 3\mathcal{E}_M) \cap D_2.$$

The implementation tests the inner sets for emptiness and plans against the outer sets. The exact set used in the statistical proof satisfies

$$\hat{K}_M(x) \subseteq \hat{K}_M^{\text{in}}(x) \subseteq \hat{K}_M^{\text{out}}(x).$$

The constants 2 and 3 are not important. Their role, together with the two expanded outcome balls, is to create a strict margin between the set containing an uninformative empirical outcome and the set used for planning.

Write $\hat{f}(x) = Fx + f_0$. For a fixed arm, the distance between $\hat{f}(x) + 2\mathcal{E}_M$ and D_1 is

$$d_M(x) := \max_{\|u\|_2 \leq 1} \left(u^T (\hat{f}(x) - c_D) - 2\sqrt{u^T V_M u} - (R_D + \rho)\|u\|_2 \right).$$

This is the support-function formula for the distance between two closed convex sets. In particular, $d_M(x) > 0$ exactly when $\hat{K}_M^{\text{in}}(x) = \emptyset$.

After introducing nonnegative variables s_0, s_1 satisfying

$$u^T V_M u \leq s_0^2, \quad \|u\|_2^2 \leq s_1^2,$$

maximizing $d_M(x)$ jointly over $x \in X$ and $\|u\|_2 \leq 1$ becomes a QCQP with a fixed number of quadratic constraints. The only coupling between the arm and dual variables is the bilinear expression $u^T Fx$. We impose the explicit bounds $\|u\|_2 \leq 1$, $0 \leq s_1 \leq 1$, and $0 \leq s_0 \leq \sqrt{\lambda_{\max}(V_M)}$, together with the ball constraint on x , and combine them into a redundant positive-definite enclosing ellipsoid. The weak fixed-constraint result in Section A then applies.

A weakly feasible output is corrected before it is used as a certificate: project its x and u components onto their respective balls, reset s_0, s_1 to the corresponding exact norms, and directly evaluate the support-function objective. Since all variables are polynomially bounded, using sufficiently smaller internal accuracy makes the change in objective at most ζ . The corrected point is feasible, so a positive value is a genuine certificate of emptiness.

Exact emptiness classification is unnecessary. Let $\zeta < \frac{\min(\rho, \lambda)}{10}$ and solve the preceding maximization to additive accuracy ζ . If an arm has certified distance greater than 2ζ , its inner set is genuinely empty. Otherwise every inner pair of constituent sets is at distance at most 3ζ . The corresponding outer intersection then contains a Euclidean ball of radius at least

$$\min(\lambda, \rho - 3\zeta).$$

Indeed, the positive semidefiniteness of $V_M - \lambda^2 I_{d_D}$ implies that $\mathcal{E}_M$ contains the Euclidean ball of radius λ . The gap from $2\mathcal{E}_M$ to $3\mathcal{E}_M$ therefore supplies the ellipsoid margin, while the gap from D_1 to D_2 supplies the outcome-ball margin.

3.1.2. Robust Planning

The preceding strict-feasibility margin also makes robust planning tractable. Fenchel–Rockafellar duality gives the following dual representation (see [7]):

$$\min_{y \in \hat{K}_M^{\text{out}}(x)} r(x, y) = a^T x + c + b^T c_D + \max_{u \in \mathbb{R}^{d_D}} \left(u^T (\hat{f}(x) - c_D) - 3\sqrt{u^T V_M u} - (R_D + 2\rho)\|b - u\|_2 \right).$$

The interior ball bounds the norm of an optimal dual vector by a polynomial in the bit length of the public and accumulated numerical data, T , and $\min(\lambda, \rho - 3\zeta)^{-1}$. Introducing epigraph variables for the two square roots therefore turns the joint maximization over x and u into a bounded QCQP with a fixed number of quadratic constraints. Appendix A gives an additive- ζ_v optimal arm in polynomial time. As in the emptiness problem, a weakly feasible solution is projected onto the arm ball and its epigraph variables are conservatively increased. The resulting loss is absorbed into ζ_v .

3.1.3. Finite-Precision Comparisons

The remaining comparisons use the same kind of gray zone. Let $\tilde{y}$ denote the rounded block average used by the implementation, and choose the rounding precision τ so that $\|\tilde{y} - \bar{y}\|_2 \leq \tau \leq \rho$. Define the residual used in the leverage test to be $\hat{g} := \tilde{y} - \hat{f}(x)$. Since $\bar{y} \in D$, the expanded ball D_1 absorbs the rounding error and $\tilde{y} \in D_1$.

Let $\ell := \hat{g}^T V_M^{-1} \hat{g}$ and compute an approximation $\tilde{\ell}$ satisfying $|\tilde{\ell} - \ell| \leq \frac{\gamma}{2}$. Append the residual only if $\tilde{\ell} - \frac{\gamma}{2} > 1$. Such an update has $\ell > 1$ and therefore retains the determinant-doubling argument in Lemma 7. Otherwise $\ell \leq 1 + \gamma$, so, for $\gamma \leq 1$,

$$\hat{g} \in \sqrt{1 + \gamma} \mathcal{E}_M \subseteq 2\mathcal{E}_M.$$

Finally, the proof of Lemma 8 is unchanged when $\mathcal{E}_M$ is multiplied by any fixed constant. Thus points in the outer ellipsoid remain $\tilde{O}(n^{-\frac{1}{2}})$ -close to $f(x) + N$, while points in D_2 are at distance at most 2ρ from D . Applying Lemma 9 shows that every point in an outer fiber is $\tilde{O}(n^{-\frac{1}{2}})$ -close to the corresponding true feasible set.

It remains to connect this numerical wrapper to the statistical proof. If a complete block is not appended, its rounded residual lies in $2\mathcal{E}_M$, and therefore

$$\tilde{y} \in \hat{K}_M^{\text{in}}(x) \subseteq \hat{K}_M^{\text{out}}(x).$$

Consequently, a complete block selected using a certified empty inner fiber must be appended; otherwise the displayed membership would contradict emptiness. Every complete block that is not appended is therefore a planning block. For such a block, define

$$\hat{v}_M^{\text{out}}(x') := \min_{y \in \hat{K}_M^{\text{out}}(x')} r(x', y).$$

The preceding outer-fiber bound and the Lipschitz property of the reward give, uniformly in x' ,

$$\hat{v}_M^{\text{out}}(x') \geq v^*(x') - \tilde{O}(n^{-\frac{1}{2}}).$$

If the weak planner returns an additive- ζ_v maximizer and x^* maximizes v^* , then

$$r(x, \bar{y}) \geq r(x, \tilde{y}) - \|b\|_2 \tau \geq \hat{v}_M^{\text{out}}(x) - \|b\|_2 \tau \geq \hat{v}_M^{\text{out}}(x^*) - \zeta_v - \|b\|_2 \tau \geq V^* - \tilde{O}(n^{-\frac{1}{2}}) - \zeta_v - \|b\|_2 \tau.$$

Affineness again identifies $r(x, \bar{y})$ with the realized average reward in the block. Thus certified empty-fiber blocks and leverage updates are charged to the same $O(\log T)$ informative blocks as before, while every other complete block has average regret at most $\tilde{O}(n^{-\frac{1}{2}}) + \zeta_v + \|b\|_2 \tau$.

Taking $\rho = \lambda = \tilde{O}(n^{-\frac{1}{2}})$ and choosing, for example, $\zeta = (T + 1)^{-8}$, $\zeta_v = (T + 1)^{-3}$, and $\tau = (T + 1)^{-12}$ makes every numerical error negligible or absorbs it into the existing $\tilde{O}(n^{-\frac{1}{2}})$ bound. These choices require only $O(\log T)$ additional precision bits, and the planning error contributes at most $T\zeta_v$ to regret. Hence the weak bit-model implementation remains polynomial in L_{pub} and T and has the same regret rate.

4. NP-hardness

In this section we show that several natural variations of Problem 1 are NP-hard. Thus in some sense, the problem is the hardest problem that's still tractable.

In this section, a **rational instance of bit length L** has a finite binary description of every set, reward coefficient, and constraint map explicitly presented in the relevant result, of total bit length L . When a fixed hypothesis is used for known-hypothesis planning, its rational description is also included in L . This full description is available to the reduction, but a learner invoked by the reduction still receives

only the corresponding public description of (X, D, r) . For each variation below, that public description uses the set representation stated in the result, and the learner's running time is polynomial in its length and T .

We first show that efficient planning can be reduced to efficient learning. Most proofs below show that planning is already NP-hard. Due to the reduction from planning to learning, this shows that learning is also NP-hard.

Lemma 11. *Fix a rational imprecise-bandit instance of bit length L and a rational hypothesis z such that $K_z(x)$ is nonempty for every $x \in X$, and write*

$$v_z(x) := \min_{y \in K_z(x)} r(x, y), \quad V_z := \max_{x \in X} v_z(x).$$

Suppose that, for every rational arm x produced by the learner, one can compute an exact rational outcome

$$y_z(x) \in \operatorname{argmin}_{y \in K_z(x)} r(x, y)$$

in time polynomial in L and the encoding length of x . Suppose also that, for some fixed $\beta > 0$ and polynomial P , a polynomial-time learner guarantees

$$E[R_T] \leq P(L)T^{1-\beta}$$

against every compatible nature policy. Then, for every $\varepsilon \in (0, 1)$, one can compute in time $\operatorname{poly}(L, \frac{1}{\varepsilon})$ an arm $\hat{x}$ and its value $\hat{V} := v_z(\hat{x})$ such that

$$\Pr(0 \leq V_z - \hat{V} \leq \varepsilon) \geq \frac{2}{3}.$$

If the learner is deterministic, the resulting planner satisfies $0 \leq V_z - \hat{V} \leq \varepsilon$ with certainty.

Proof. Run the learner on the public description of (X, D, r) for

$$T := \left\lceil \left(3 \frac{P(L)}{\varepsilon} \right)^{\frac{1}{\beta}} \right\rceil$$

rounds with true hypothesis z , and let nature put a point mass on $y_z(x_t)$ after each arm x_t . Choose

$$\hat{t} \in \operatorname{argmax}_{t=1, \dots, T} v_z(x_t)$$

and return $\hat{x} := x_{\hat{t}}$ and $\hat{V} := v_z(\hat{x})$.

Since nature always returns a minimizing outcome,

$$0 \leq V_z - \hat{V} \leq \frac{1}{T} \sum_{t=1}^T (V_z - v_z(x_t)) = \frac{R_T}{T}.$$

Consequently,

$$E[V_z - \hat{V}] \leq P(L)T^{-\beta} \leq \frac{\varepsilon}{3}.$$

Markov's inequality gives the claimed probability. The horizon and the simulation time are polynomial in L and $\frac{1}{\varepsilon}$ because β is fixed. If the learner is deterministic, then the simulation is deterministic and the displayed bound holds without taking an expectation. ■

4.1. $D = \text{simplex}$, $X = \text{ball}$, $F = F_0 + F_1$

We first replace the Euclidean outcome ball in Problem 1 by a simplex. Even known-hypothesis planning then becomes NP-hard.

Theorem 2. *There is a polynomial-time reduction from MAX-CUT to rational imprecise-bandit instances for which X is a Euclidean unit ball, D is a simplex, $Z = [1, 2]$, the maps F_0 and F_1 are bilinear, and the reward is linear and takes values in $[0, 1]$. Every $K_z(x)$ is nonempty, and every $z \in Z$ induces the same model.*

For these instances, approximating the known-hypothesis value

$$V_z := \max_{x \in X} \min_{y \in K_z(x)} r(x, y)$$

to inverse-polynomial additive accuracy is NP-hard. Consequently, if, for some fixed $\beta > 0$, a randomized polynomial-time learner guaranteed

$$E[R_T] \leq P(L)T^{1-\beta}$$

on every such instance, where L is the total rational-description length and P is a polynomial, then $\text{NP} \subseteq \text{BPP}$. For a deterministic learner, the same conclusion would be $\text{P} = \text{NP}$.

Proof. We reduce from MAX-CUT [8]. Let $G = (V, E)$ have n vertices and $m \geq 1$ edges. Orient the edges arbitrarily, let $B_G \in \mathbb{R}^{m \times n}$ be the resulting edge-vertex incidence matrix, and put

$$\gamma := \frac{1}{2mn}, \quad X := \{x \in \mathbb{R}^m : \|x\|_2 \leq 1\}, \quad Z := [1, 2].$$

Write an outcome as $y = (p, q, s) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$ and take

$$D := \left\{ (p, q, s) : p_i \geq 0, q_i \geq 0 \text{ for } i = 1, \dots, n, s \geq 0, \sum_{i=1}^n (p_i + q_i) + s = 1 \right\}.$$

With $W = \mathbb{R}^n$, define

$$F_0(x, z) := -\gamma z B_G^T x, \quad F_1((p, q, s), z) := z(p - q),$$

and

$$r(x, (p, q, s)) := \sum_{i=1}^n (p_i + q_i).$$

Both constraint maps are bilinear, and the reward lies in $[0, 1]$ on D . Since $z \neq 0$, compatibility is equivalent to

$$p - q = \gamma B_G^T x,$$

so it is independent of z .

The incidence matrix has $\|B_G\|_F = \sqrt{2m}$. Hence every $x \in X$ satisfies

$$\|B_G^T x\|_1 \leq \sqrt{n} \|B_G^T x\|_2 \leq \sqrt{n} \|B_G\|_F \|x\|_2 \leq \sqrt{2mn}.$$

For $t := \gamma B_G^T x$, it follows that $\|t\|_1 \leq 1$. The choice

$$p_i := \max(t_i, 0), \quad q_i := \max(-t_i, 0), \quad s := 1 - \|t\|_1$$

therefore belongs to D and satisfies $p - q = t$, proving that every $K_z(x)$ is nonempty.

For any compatible (p, q, s) , nonnegativity gives $p_i + q_i \geq |p_i - q_i| = |t_i|$. The preceding positive-negative decomposition attains equality in every coordinate, and therefore

$$v_z(x) := \min_{y \in K_z(x)} r(x, y) = \gamma \|B_G^T x\|_1.$$

By duality of the ℓ_1 and ℓ_∞ norms,

$$\|B_G^T x\|_1 = \max_{\sigma \in \{-1, 1\}^n} \sigma^T B_G^T x.$$

Consequently,

$$\max_{x \in X} \|B_G^T x\|_1 = \max_{\sigma \in \{-1, 1\}^n} \|B_G \sigma\|_2.$$

For a sign vector σ and an edge $e = \{i, j\}$, the corresponding coordinate satisfies

$$(B_G \sigma)_e^2 = (\sigma_i - \sigma_j)^2 = \begin{cases} 4 & \text{if } \sigma_i \neq \sigma_j \\ 0 & \text{otherwise} \end{cases}.$$

Thus $\|B_G \sigma\|_2^2$ is four times the size of the cut encoded by σ . If M_G denotes the maximum cut size, then

$$V_z = 2\gamma \sqrt{M_G}.$$

The values corresponding to consecutive cut sizes satisfy, for $k = 1, \dots, m$,

$$2\gamma(\sqrt{k} - \sqrt{k-1}) = 2\frac{\gamma}{\sqrt{k} + \sqrt{k-1}} \geq \frac{\gamma}{m}.$$

The tolerance $\frac{\gamma}{3m}$ is inverse-polynomial in the encoding length. Approximating the candidate values $2\gamma\sqrt{k}$ to polynomially many bits, an estimate of V_z within this tolerance identifies M_G . This proves the planning claim.

The positive-negative decomposition displayed above is an exact rational minimizing outcome and is computable in polynomial time from every rational arm. Applying Lemma 11 with $\varepsilon = \frac{\gamma}{3m}$ therefore supplies, with probability at least $\frac{2}{3}$, the estimate used in the preceding paragraph. This would place MAX-CUT in BPP and hence imply $\text{NP} \subseteq \text{BPP}$. For a deterministic learner, the estimate is deterministic and gives $\text{P} = \text{NP}$. ■

4.2. $D = \text{ball}$, $X = \text{polytope}$, $F = F_0 + F_1$

We now keep the Euclidean outcome ball and the additive bilinear constraint, but allow the arm set to be a polytope given by rational inequalities. Even the cube, which has only $2n$ such inequalities but 2^n vertices, makes known-hypothesis planning NP-hard.

Theorem 3. *There is a polynomial-time reduction from MAX-CUT to rational imprecise-bandit instances for which $X = [-1, 1]^n$, represented by $2n$ inequalities, D is a Euclidean unit ball, $Z = [1, 2]$, the maps F_0 and F_1 are bilinear, and the reward is linear. Every $K_z(x)$ is nonempty, every $z \in Z$ induces the same model, and the uniform non-tangency condition holds with an absolute constant.*

For these instances, approximating the known-hypothesis value

$$V_z := \max_{x \in X} \min_{y \in K_z(x)} r(x, y)$$

to inverse-polynomial additive accuracy is NP-hard. Consequently, if, for some fixed $\beta > 0$, a randomized polynomial-time learner guaranteed

$$E[R_T] \leq P(L)T^{1-\beta}$$

on every such instance, where L is the total rational-description length and P is a polynomial, then $\text{NP} \subseteq \text{BPP}$. For a deterministic learner, the same conclusion would be $\text{P} = \text{NP}$.

Proof. We again reduce from MAX-CUT. Let $G = (V, E)$ have n vertices and $m \geq 1$ edges. Orient the edges arbitrarily and let $B_G \in \mathbb{R}^{m \times n}$ be the edge-vertex incidence matrix, so that

$$(B_G x)_e = x_i - x_j$$

for an edge $e = \{i, j\}$. Put

$$\varepsilon := \frac{1}{4m}, \quad X := [-1, 1]^n, \quad Z := [1, 2].$$

Write outcomes as $y = (u, s) \in \mathbb{R}^m \times \mathbb{R}$ and take

$$D := \{(u, s) : \|u\|_2^2 + s^2 \leq 1\}.$$

With $W = \mathbb{R}^m$, define

$$F_0(x, z) := -\varepsilon z B_G x, \quad F_1((u, s), z) := zu,$$

and let $r(x, (u, s)) := s$. Both constraint maps are bilinear and the reward is linear. Since $z \neq 0$, compatibility is equivalent to

$$u = \varepsilon B_G x,$$

and is therefore independent of z . Moreover, every $x \in X$ satisfies

$$\|B_G x\|_2^2 = \sum_{\{i, j\} \in E} (x_i - x_j)^2 \leq 4m,$$

so

$$\|\varepsilon B_G x\|_2^2 \leq \frac{1}{4m} \leq \frac{1}{4}.$$

Thus every compatible fiber is nonempty. Minimizing s over that fiber gives

$$v_z(x) = -\sqrt{1 - \varepsilon^2 \|B_G x\|_2^2}.$$

Set $q(x) := \|B_G x\|_2^2$. With every coordinate except x_i fixed, $q(x)$ is a convex quadratic in x_i , so replacing x_i by one of the endpoints of $[-1, 1]$ does not decrease q . Applying this replacement one coordinate at a time produces a sign vector $\sigma \in \{-1, 1\}^n$ with $q(\sigma) \geq q(x)$. At a sign vector,

$$q(\sigma) = \sum_{\{i,j\} \in E} (\sigma_i - \sigma_j)^2 = 4 |\text{cut}(\sigma)|.$$

Consequently,

$$\max_{x \in [-1, 1]^n} q(x) = 4 \text{MaxCut}(G).$$

The displayed robust value is strictly increasing in $q(x)$. If the maximum cut has size k , the planning value is therefore

$$V_k = -\sqrt{1 - \frac{k}{4m^2}}.$$

For $k = 1, \dots, m$, consecutive possible values satisfy

$$V_k - V_{k-1} = \frac{\frac{1}{4m^2}}{\sqrt{1 - \frac{k-1}{4m^2}} + \sqrt{1 - \frac{k}{4m^2}}} \geq \frac{1}{8m^2}.$$

Hence an estimate of V_z within

$$\Delta := \frac{1}{32m^2}$$

identifies k after the $m + 1$ candidate values are computed to polynomially many bits. This proves the planning claim.

We next derive the learning consequence. The exact minimizing outcome displayed above need not be rational, so we use a rational feasible approximation rather than invoke Lemma 11. Suppose the claimed learner exists and set

$$\eta := \frac{\Delta}{6}, \quad T := \left\lceil \left(6 \frac{P(L)}{\Delta} \right)^{\frac{1}{\beta}} \right\rceil.$$

In the weak bit model the learner outputs rationally encoded arms. For each played arm x_t , rational binary search for a square root computes a_t in polynomial time such that

$$0 \leq \sqrt{1 - \varepsilon^2 q(x_t)} - a_t \leq \eta, \quad a_t^2 \leq 1 - \varepsilon^2 q(x_t).$$

Let nature return the point

$$y_t := (\varepsilon B_G x_t, -a_t).$$

This is a rational point of $K_z(x_t)$, and its reward satisfies

$$0 \leq r(x_t, y_t) - v_z(x_t) \leq \eta.$$

Writing $V_z = V_k$ and

$$S_T := \sum_{t=1}^T (V_z - v_z(x_t)),$$

we therefore have

$$S_T \leq R_T + T\eta.$$

Taking expectations under the learner's randomization gives

$$E[S_T] \leq P(L)T^{1-\beta} + T\eta.$$

If the learner never plays a Δ -optimal arm, then $S_T > \Delta T$. Markov's inequality and the choices of T and η imply

$$\Pr(\text{no } \Delta\text{-optimal arm is played}) \leq \frac{P(L)}{\Delta T^\beta} + \frac{\eta}{\Delta} \leq \frac{1}{3}.$$

Select a played arm maximizing the rational quantity $q(x_t)$ and round its coordinates to endpoints without decreasing q . With probability at least $\frac{2}{3}$, the resulting sign vector has robust value at least $V_k - \Delta$. A cut of size at most $k - 1$ would instead have robust value at most $V_{k-1} \leq V_k - \frac{1}{8m^2} < V_k - \Delta$. The rounded sign vector therefore encodes a maximum cut. The horizon, the square-root approximations, and the rounding all have polynomial bit complexity, proving $\text{NP} \subseteq \text{BPP}$. If the learner is deterministic, the same argument succeeds with certainty and gives $\text{P} = \text{NP}$.

It remains to verify non-tangency. For a fixed x , put

$$a := \|\varepsilon B_G x\|_2, \quad \rho := \sqrt{1 - a^2} \geq \frac{\sqrt{3}}{2}.$$

The ambient affine solution space and its intersection with D are

$$L_z(x) = \{(\varepsilon B_G x, s) : s \in \mathbb{R}\}, \quad K_z(x) = \{(\varepsilon B_G x, s) : |s| \leq \rho\}.$$

If $p = (\varepsilon B_G x, s) \in L_z(x)$ lies outside D and $t := |s| > \rho$, then

$$\text{dist}(p, K_z(x)) = t - \rho, \quad \text{dist}(p, D) = \sqrt{a^2 + t^2} - 1.$$

Factoring the second expression gives

$$\frac{\text{dist}(p, D)}{\text{dist}(p, K_z(x))} = \frac{t + \rho}{\sqrt{a^2 + t^2} + 1} \geq \rho \geq \frac{\sqrt{3}}{2}.$$

Indeed, after clearing the positive denominator, subtracting ρ , and squaring, the first inequality is equivalent to $(1 - \rho^2)(t^2 - \rho^2) \geq 0$. Thus the uniform non-tangency condition holds with $S = \frac{\sqrt{3}}{2}$. ■

4.3. $D = \text{ball}$, $X = \text{ball}$, $F = \text{trilinear}$

In this subsection, we retain the Euclidean arm and outcome balls from Problem 1 but drop the additive restriction and allow a general trilinear constraint map. It is convenient to use the homogeneous outcome formulation. Thus, if Y is the ambient outcome space, W is the constraint space, and

$$F : \mathbb{R}^{d_X} \times \mathbb{R}^{d_Z} \times Y \rightarrow W$$

is linear in each argument separately, define

$$L_z(x) := \ker F_{x,z}, \quad K_z(x) := L_z(x) \cap D, \quad v_z(x) := \min_{y \in K_z(x)} r(x, y), \quad V_z := \max_{x \in X} v_z(x).$$

Here D is a Euclidean ball in the affine normalization hyperplane, hence a Euclidean ball in its affine hull. The construction below shows that allowing the coefficients of the outcome constraint to depend on the arm makes even known-hypothesis planning hard.

Let $G = (V, E)$ be an undirected graph with $V = \{1, \dots, n\}$ and $E = \{e_1, \dots, e_m\}$, where $m \geq 1$ and $e_\ell = \{i_\ell, j_\ell\}$. Put $d := n + m$. For $q = (u, w) \in \mathbb{R}^n \times \mathbb{R}^m$, define the homogeneous cubic

$$p_G(q) := \sum_{\ell=1}^m u_{i_\ell} u_{j_\ell} w_\ell.$$

Lemma 12. *If $\omega(G)$ is the clique number of G , then*

$$\max_{\|q\|_2=1} p_G(q)^2 = \frac{2}{27} \left(1 - \frac{1}{\omega(G)}\right).$$

Proof. Write $\rho := \|u\|_2$ and $\sigma := \|w\|_2$. For fixed u and σ , Cauchy–Schwarz gives

$$\max_{\|w\|_2=\sigma} p_G(u, w) = \sigma \sqrt{\sum_{\ell=1}^m u_{i_\ell}^2 u_{j_\ell}^2}.$$

Writing $u = \rho s$ with $\|s\|_2 = 1$ and setting $\pi_i := s_i^2$, the Motzkin–Straus theorem [9] gives

$$\max_{\|s\|_2=1} \sum_{\{i,j\} \in E} s_i^2 s_j^2 = \max_{\pi_i \geq 0, \sum_i \pi_i = 1} \sum_{\{i,j\} \in E} \pi_i \pi_j = \frac{1}{2} \left(1 - \frac{1}{\omega(G)}\right).$$

Finally,

$$\max_{\rho^2 + \sigma^2 = 1, \rho \geq 0, \sigma \geq 0} \rho^2 \sigma = \frac{2}{3\sqrt{3}}.$$

Squaring the product of these two maxima proves the identity. Since p_G is odd, its maximum, rather than only its maximum absolute value, is the positive square root of the displayed quantity. ■

Theorem 4. *There is a polynomial-time reduction from CLIQUE to rational imprecise-bandit instances for which X is a full-dimensional Euclidean ball, D is a Euclidean ball in its affine hull, $Z = [1, 2]$ is a one-dimensional Euclidean ball, the reward is linear, and F is trilinear. Every set $K_z(x)$ is a singleton independent of z , and the uniform non-tangency condition holds. Even the corresponding distance inequality over the full homogeneous kernel holds with a constant $S > 0.88$.*

For these instances, approximating V_z to inverse-polynomial additive accuracy is NP-hard even when z is known. Consequently, if, for some fixed $\beta > 0$, a randomized polynomial-time learner guaranteed

$$E[R_T] \leq P(L)T^{1-\beta}$$

on every such instance for a polynomial P , then $\text{NP} \subseteq \text{BPP}$. For a deterministic learner, the same conclusion would be $\text{P} = \text{NP}$.

Proof. We may restrict CLIQUE to instances with $m \geq 1$ and $3 \leq k \leq n$, since the excluded cases are polynomial-time decidable. Starting from (G, k) , use the cubic p_G above and write an arm as $x = (x_0, \xi) \in \mathbb{R} \times \mathbb{R}^d$. Take

$$X := \left\{ (x_0, \xi) : \left(x_0 - \frac{5}{3} \right)^2 + \|\xi\|_2^2 \leq 1 \right\}.$$

This is a full-dimensional Euclidean ball and $x_0 \geq \frac{2}{3}$ throughout X . Moreover, the normalized coordinate $q := \frac{\xi}{x_0}$ ranges over exactly the ball $\|q\|_2 \leq \frac{3}{4}$. Indeed,

$$1 - \left(x_0 - \frac{5}{3} \right)^2 - \frac{9}{16}x_0^2 = -\frac{(15x_0 - 16)^2}{144} \leq 0,$$

so every attainable ratio has norm at most $\frac{3}{4}$. Conversely, for any such q , the choice $x_0 = \frac{16}{15}$ and $\xi = x_0 q$ belongs to X .

Let $Z := [1, 2]$. Introduce the ambient outcome and constraint spaces

$$Y := \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^{d \times d} \times \mathbb{R}^{d \times d \times d},$$

$$W := \mathbb{R}^d \times \mathbb{R}^{d \times d} \times \mathbb{R}^{d \times d \times d},$$

and write $y = (y_0, \eta, \Theta, \Xi) \in Y$. With the normalization functional $\mu(y) := y_0$, the outcome set is the affine Euclidean unit ball

$$D := \{ (1, \eta, \Theta, \Xi) : \|\eta\|_2^2 + \|\Theta\|_F^2 + \|\Xi\|_F^2 \leq 1 \}.$$

For $x = (x_0, \xi)$, $z \in \mathbb{R}$, and $y \in Y$, define $F(x, z, y)$ coordinatewise by

$$F_i(x, z, y) := z \left(x_0 \eta_i - \frac{1}{2} \xi_i y_0 \right),$$

$$F_{ij}(x, z, y) := z (x_0 \Theta_{ij} - \xi_i \eta_j),$$

$$F_{ij\ell}(x, z, y) := z (x_0 \Xi_{ij\ell} - \xi_i \eta_{j\ell}).$$

Each term contains one coordinate from each of x , z , and y , so F is trilinear. Define the reward by

$$r(x, y) := \frac{1}{m} \sum_{\ell=1}^m \Xi_{i_\ell, j_\ell, n+\ell}.$$

It is linear and independent of x ; its coefficient vector has Euclidean norm $\frac{1}{\sqrt{m}} \leq 1$.

Fix $x \in X$ and $z \in Z$. Since $zx_0 \neq 0$ and $y_0 = 1$ on D , the equations $F(x, z, y) = 0$ can be solved recursively. They force the unique candidate

$$\eta_i = \frac{1}{2} q_i, \quad \Theta_{ij} = \frac{1}{2} q_i q_j, \quad \Xi_{ij\ell} = \frac{1}{2} q_i q_j q_\ell.$$

If $t := \|q\|_2 \leq \frac{3}{4}$, the squared norm of its nonconstant outcome coordinates is

$$\frac{1}{4} (t^2 + t^4 + t^6) \leq \frac{4329}{16384} < 1.$$

Thus the candidate lies in D , proving that $K_z(x)$ is precisely this singleton. It does not depend on z . The operator $F_{x,z} : Y \rightarrow W$ is also onto: set $y_0 = 0$ and solve successively for η , Θ , and Ξ for an arbitrary right-hand side.

On the normalization hyperplane, the affine solution space is the singleton $K_z(x)$, so the intrinsic non-tangency condition in Problem 1 is automatic. In fact, a stronger homogeneous version holds. Surjectivity

and the dimensions of Y and W show that $L_z(x)$ is the line spanned by $y(x) = (1, g(q))$, where $g(q) := (\eta, \Theta, \Xi)$ is given by the displayed recursion. If $p = \tau y(x) \in L_z(x)$ lies outside D , then

$$\text{dist}(p, D) \geq |\tau - 1|, \quad \text{dist}(p, K_z(x)) = |\tau - 1| \sqrt{1 + \|g(q)\|_2^2}.$$

Consequently the ambient homogeneous distance inequality holds uniformly with

$$S \geq \frac{1}{\sqrt{1 + \frac{4329}{16384}}} = \frac{128}{\sqrt{20713}} > 0.88.$$

Because the compatible outcome is unique, evaluating the reward at that outcome gives

$$v_z(x) = \frac{1}{2m} p_G(q).$$

The ratio q fills the radius- $\frac{3}{4}$ ball, so homogeneity and Lemma 12 imply, for every $z \in Z$,

$$V_z = \frac{27}{128m} \max_{\|h\|_2=1} p_G(h) = \frac{27}{128m} \sqrt{\frac{2}{27}} \sqrt{1 - \frac{1}{\omega(G)}}.$$

For $j = 2, \dots, n$, let

$$U_j := \frac{27}{128m} \sqrt{\frac{2}{27}} \sqrt{1 - \frac{1}{j}}.$$

If $\omega(G) \geq k$, then $V_z \geq U_k$; if $\omega(G) \leq k-1$, then $V_z \leq U_{k-1}$. The two cases have gap

$$U_k - U_{k-1} \geq \frac{27}{128m} \sqrt{\frac{2}{27}} \frac{1}{2k(k-1)} = \Omega\left(\frac{1}{mk^2}\right).$$

Since $\sqrt{\frac{2}{27}} > \frac{1}{4}$ and $m \leq n^2$, the gap $\Delta_k := U_k - U_{k-1}$ satisfies

$$\Delta_k > \frac{1}{40mk(k-1)} \geq \frac{1}{40mn^2}.$$

Set $\varepsilon := \frac{1}{500mn^2}$. Using polynomially many bits, one can compute a rational τ_k such that

$$\left| \tau_k - \frac{1}{2}(U_k + U_{k-1}) \right| \leq \varepsilon.$$

An ε -additive estimate of V_z lies above τ_k when $\omega(G) \geq k$ and below it when $\omega(G) \leq k-1$, because $2\varepsilon < \frac{\Delta_k}{2}$. It therefore decides whether G contains a k -clique. The constructed instance has dimension $O(d^3)$, and all its defining coefficients are rational with polynomial encoding length. This proves the planning claim.

The displayed recursion computes the unique compatible outcome exactly in polynomial time from every rational arm; in particular, $x_0 \geq \frac{2}{3}$ keeps the division $q = \frac{\xi}{x_0}$ bounded. Applying Lemma 11 with the displayed ε therefore supplies, with probability at least $\frac{2}{3}$, an additive estimate to which the preceding threshold argument applies. This would place CLIQUE in BPP and hence imply $\text{NP} \subseteq \text{BPP}$. For a deterministic learner, the estimate is deterministic and gives $\text{P} = \text{NP}$. ■

4.4. IUCB is NP-hard even when X and D are Euclidean balls

Given rational descriptions of X , D , Z , the affine reward r , and the maps defining $K_z(x)$, let

$$v_z(x) := \min_{y \in K_z(x)} r(x, y).$$

The **exact IUCB optimism problem** asks for the value

$$V_{\text{IUCB}} := \max_{z \in Z, x \in X} v_z(x).$$

This is the optimization performed on the first round of IUCB, when its confidence set is all of Z . The associated search problem asks for a globally optimal first arm

$$x^* \in \operatorname{argmax}_{x \in X} \max_{z \in Z} v_z(x).$$

Theorem 5. *Even when X , D , and Z are Euclidean balls and planning for every fixed hypothesis is polynomial-time, computing the exact first optimistic value or a globally optimal first arm of IUCB is NP-hard.*

Proof. We reduce from the exact tensor spectral-norm problem. Given a rational order-three tensor $\mathcal{T} \in \mathbb{Q}^{m \times d \times d_X}$, viewed as a bilinear map $\mathcal{T} : \mathbb{R}^d \times \mathbb{R}^{d_X} \rightarrow \mathbb{R}^m$, this problem asks to compute

$$\|\mathcal{T}\|_\sigma := \max_{\|u\|_2 \leq 1, \|x\|_2 \leq 1} \|\mathcal{T}(u, x)\|_2.$$

Computing this value is NP-hard [10]. From $\mathcal{T}$, construct an IUCB instance as follows. Put

$$H := 1 + \sum_{i,j,k} |T_{ijk}|, \quad \alpha := \frac{1}{2H}, \quad X := \{x : \|x\|_2 \leq 1\}.$$

Write $z = (z_0, u)$ and $y = (w, s)$, and take

$$Z := \left\{ (z_0, u) : \left(z_0 - \frac{5}{3} \right)^2 + \|u\|_2^2 \leq 1 \right\}, \quad D := \{(w, s) : \|w\|_2^2 + s^2 \leq 1\}.$$

For $z = (z_0, u)$, define the data in Equation 1 and the reward by

$$B_z x := -\alpha \mathcal{T}(u, x), \quad C_z(w, s) := z_0 w, \quad d_z := 0, \quad r(x, (w, s)) := \frac{1}{2}(1 + s).$$

These are instances of Problem 1, and

$$K_z(x) = \left\{ (w, s) \in D : w = \frac{\alpha}{z_0} \mathcal{T}(u, x) \right\}.$$

The definition of Z gives $z_0 \geq \frac{2}{3}$ and

$$\frac{\|u\|_2}{z_0} \leq \frac{3}{4},$$

because $1 - \left(z_0 - \frac{5}{3}\right)^2 - 9\frac{z_0^2}{16} = -\frac{(15z_0-16)^2}{144} \leq 0$. Equality holds at $z_0 = \frac{16}{15}$ and $\|u\|_2 = \frac{4}{5}$ in every direction. Also $\|\mathcal{T}(u, x)\|_2 \leq H\|u\|_2\|x\|_2$, so the fixed w in $K_z(x)$ has norm at most $3\alpha\frac{H}{4} = \frac{3}{8}$. Thus every $K_z(x)$ is nonempty, and the uniform non-tangency condition holds with $S \geq \frac{\sqrt{55}}{8}$. For this instance, nature chooses $s = -\sqrt{1 - \|w\|_2^2}$ when evaluating $v_z(x)$, whence

$$v_z(x) = h\left(\frac{\alpha}{z_0} \|\mathcal{T}(u, x)\|_2\right), \quad h(t) := \frac{1}{2}(1 - \sqrt{1 - t^2}).$$

For fixed z , a top right singular vector of $M_u x := \mathcal{T}(u, x)$ maximizes $v_z(x)$, so fixed-hypothesis planning is polynomial-time. IUCB instead optimizes jointly over Z and X . Since h is strictly increasing and the ratio above attains $\frac{3}{4}$ in every direction,

$$V_{\text{IUCB}} = h\left(3\frac{\alpha}{4}\|\mathcal{T}\|_\sigma\right),$$

Since

$$\|\mathcal{T}\|_\sigma = \frac{4}{3\alpha} \sqrt{1 - (1 - 2V_{\text{IUCB}})^2},$$

exact IUCB optimism would solve the tensor spectral-norm problem. If IUCB returns only a globally optimal arm x^* , one SVD of $N_x u := \mathcal{T}(u, x^*)$ computes

$$\max_{z \in Z} v_z(x^*) = h\left(3\frac{\alpha}{4}\sigma_{\max}(N_x)\right) = V_{\text{IUCB}},$$

so the same inverse recovers $\|\mathcal{T}\|_\sigma$. Producing a globally optimal first arm is therefore NP-hard as well. ■

A. Quadratically constrained quadratic programming

For symmetric matrices $Q_i \in \mathbb{R}^{d \times d}$, a **quadratically constrained quadratic program** (QCQP) in $z \in \mathbb{R}^d$ has the form

$$\min_{z \in \mathbb{R}^d} g_0(z) \quad \text{subject to} \quad g_i(z) \leq 0, \quad i = 1, \dots, m,$$

where $g_i(z) := z^T Q_i z + 2p_i^T z + r_i$; for complexity statements, all entries are rational. A quadratic equality $h(z) = 0$ is exactly the pair $h(z) \leq 0$ and $-h(z) \leq 0$ [3].

QCQP describes algebraic form, whereas convex programming describes geometry. The displayed minimization problem is a convex program when every Q_i , $i = 0, \dots, m$, is positive semidefinite and all equalities are affine [7]. Without those restrictions it can be nonconvex: general QCQP is NP-hard even when only the objective is quadratic and its Hessian has one negative eigenvalue [11].

The polynomial-time conclusion for our nonconvex instance has two steps.

1. First, fix the number m of constraints. Bienstock’s theorem states that, for rational quadratics $g_0, \dots, g_m$, if at least one constraint $g_i(z) \leq 0$ has a positive-definite quadratic part, then for every $\varepsilon \in (0, 1)$ an algorithm runs in time $\text{poly}(L, \log(\frac{1}{\varepsilon}))$, where L is the input bit length, and either proves infeasibility or returns $\hat{z}$ such that

$$g_i(\hat{z}) \leq \varepsilon, \quad i = 1, \dots, m,$$

and $g_0(\hat{z}) \leq g_0(z) + \varepsilon$ for every feasible z [3]. The underlying weak-feasibility step reduces to a fixed number of homogeneous quadratic equations on a sphere, handled by Barvinok’s method; binary search then gives the objective guarantee [3], [12].

2. Second, our planning problem is a QCQP with a fixed number of constraints, one of which is the strictly convex joint-ellipsoid constraint when $R_X, R_D > 0$. Thus it satisfies Bienstock’s hypotheses. Applying the theorem to the negative objective computes the planning maximum to weak additive accuracy ε in time $\text{poly}(L, \log(\frac{1}{\varepsilon}))$. If either radius is zero, the problem first reduces to a lower-dimensional instance.

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